

Given Specifications:-

$$A_o > 55 \text{ dB}$$

$$\text{CMRR} > 60 \text{ dB}$$

$$\text{GBW} > 3 \text{ MHz}$$

$$\text{Phase Margin (PM)} > 45^\circ$$

$$\text{Output Swing} > 1.2 \text{ V}_{pp}$$

$$\text{Load Cap} = 30 \text{ pF}$$

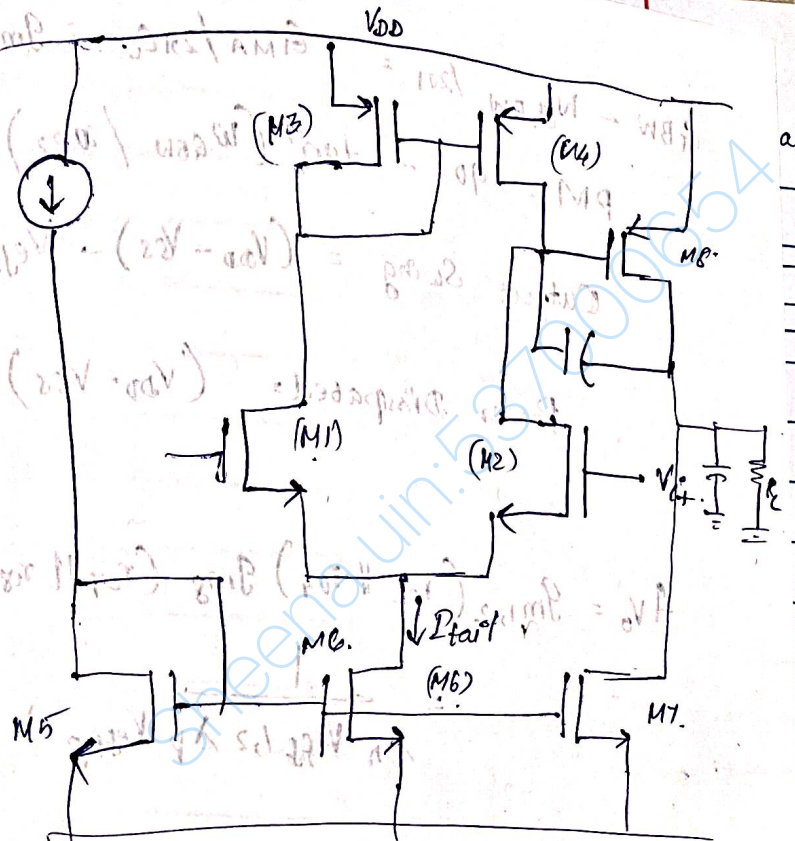
$$\text{Load Resistor} = \infty$$

$$\text{Power Dissipation} < 500 \mu\text{W}$$

$$\text{Supply} = 1.8 \text{ V}$$

Solution

$$1) I_{\text{total}} < \frac{500 \mu\text{W}}{1.8 \text{ V}} = 277 \mu\text{A}$$



$$\rightarrow G_{mA} = g_{m1,2}$$

$$G_{mB} = g_{m8}$$

$$R_A = r_{o2} \parallel r_{o4}$$

$$R_B = r_{o1} \parallel r_{o8} \parallel R_L$$

$$R_{out} = R_B$$

$$G_m = g_{m1,2} (r_{o2} \parallel r_{o4}) g_{m8}$$

$$A_{V_o} = A_{DM} = G_m R_{out} = g_{m1,2} (r_{o2} \parallel r_{o4}) g_{m8} (r_{o1} \parallel r_{o8} \parallel R_L)$$

$$A_{CM} = \frac{(2g_{m2} r_{o2} r_{o6} + r_{o2} + 2r_{o6}) \parallel r_{o4} \parallel (1/g_{m4})}{1/g_{m1,2} + 2r_{o6}} g_{m8} (r_{o1} \parallel r_{o8} \parallel R_L)$$

$$\text{CMRR} = \left| \frac{A_{DM}}{A_{CM}} \right| = \frac{(1 + 2r_{o6} g_{m1,2}) (r_{o2} \parallel r_{o4})}{(2g_{m2} r_{o2} r_{o6} + r_{o2} + 2r_{o6}) \parallel r_{o4} \parallel 1/g_{m4}}$$

$$W_{GBW} = G_{mA} / C_c = g_{m1,2} / C_c \quad W_z = G_{mB} / C_c = g_{m8} / C_c$$

$$W_{P1} = 1 / (G_{mB} R_B C_c R_A)$$

$$W_z = G_{mB} / C_c = g_{m8} / C_c$$

$$W_{P2} = G_{mB} / C_B = g_{m8} / C_L$$

$$(a) \text{ Output Swing} = 1.8 \text{ V} - V_{eff7} - V_{eff8} > 1.2 \text{ V}.$$

$$\text{Power Dissipated} = 1.8 \times (I_{bias} + I_{tail} + I_{out}) < 500 \mu\text{W}$$

$$V_x = -(V_{tn} + V_{eff1,2})$$

$$\text{Power Dissipated} = 1.8 \times (I_{bias} + I_{tail} + I_{out}) < 500 \mu\text{W}$$

$$V_x = -(V_{tn} + V_{eff1,2}) > V_{eff6} + V_{ss}.$$

$$V_{eff1,2} + V_{eff7} < 0.9 \text{ V} - V_{tn}.$$

from Lab 1, we have, $k_p n = 210 \mu\text{A/V}^2$
 $k_p p = 50 \mu\text{A/V}^2$

$$V_{tn} = 450 \text{ mV}$$

$$\lambda_n = 160 \text{ mV}^{-1}$$

$$V_{tp} = -420 \text{ mV}$$

$$\lambda_p = 190 \text{ mV}^{-1}$$

~~from Lab 1~~, $C_c = 15 \text{ pF}$

$$V_{eff1,2} = 392.58 \text{ mV}.$$

$$V_{eff3,4} = 392.58 \text{ mV}.$$

$$V_{eff5,7} = 57 \text{ mV}.$$

$$V_{eff6,8} = 121.29 \text{ mV}.$$

$$V_{eff8} = 93.10 \text{ mV}.$$

$$V_{eff} = 70 \text{ mV}.$$

from equation.

$$1.8 \times (I_{bias} + I_{tail} + I_{out})$$

$$1.8 \times (I_{bias} + 4I_{bias} + 2I_{bias}) < 500 \mu\text{W}$$

$$7 \times 1.8 I_{bias} < 500 \mu\text{W}$$

$$I_{bias} < 39.68 \mu\text{A}$$

$$I_{bias} \leq 37 \mu\text{A}$$

$$I_{tail} \leq 4(37) \mu\text{A}$$

$$I_{tail} = 148 \mu\text{A}$$

Finding $V_{eff} = V_{GS} - V_t \Rightarrow V_{GS} = V_t + V_{eff}$

$V_{eff\ 1,2} = 342.58\text{ mV}$

$V_{GS\ 1,2} = 842.58\text{ mV}$

$V_{GS\ 1,2} = 839.842\text{ mV}$

(n) $V_{eff\ 5,7} = 57\text{ mV}$

$V_{GS\ 5,7} = 507\text{ mV}$

$V_{DS\ 5,7} = 507\text{ mV}$

(m) $V_{eff\ 3,4} = 121.29\text{ mV}$

$V_{GS\ 3,4} = 541.29\text{ mV}$

$V_{SD\ 3,4} = 513.1$

(p) $V_{eff\ 8} = 93.10\text{ mV}$

$V_{GS\ 8} = 513.1$

$V_{DS\ 6} = 130\text{ mV}$

(p) $V_{eff\ 6} = 70\text{ mV}$

$V_{GS\ 6} = 520\text{ mV}$

$V_{DS\ 7} = 900\text{ mV}$

(n) $V_{eff\ 6} = 70\text{ mV}$

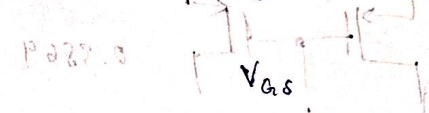
$V_{GS\ 6} = 520\text{ mV}$

$V_{DS\ 8} = 900\text{ mV}$

Claiming $W = \frac{2 I_D L}{k_p (V_{eff})^2 (1 + \lambda(V_{DS} + V_{SD}))}$

$k_p (V_{eff})^2 (1 + \lambda(V_{DS} + V_{SD}))$

Finding W/L



$M_{1,2}$

842.58

$V_{DS/SD}$

839.84 mV

I_D

$148\text{ }\mu\text{A}$

W/L

$1.2\text{ }\mu\text{m} / 0.27\text{ }\mu\text{m}$

$M_{3,4}$

541.29

513.1 mV

$74\text{ }\mu\text{A}$

$14 / 0.27$

M_5

507 mV

507 mV

$37\text{ }\mu\text{A}$

$6.5\text{ }\mu\text{m} / 0.27$

M_6

520 mV

130 mV

$148\text{ }\mu\text{A}$

$28\text{ }\mu\text{m} / 0.27$

M_7

507 mV

900 mV

$148\text{ }\mu\text{A}$

$28\text{ }\mu\text{m} / 0.27$

M_8

513.1 mV

900 mV

$148\text{ }\mu\text{A}$

$104\text{ }\mu\text{m} / 0.27$